

Retrieval-guided Twin Fusion with Similarity-aware Contrast for Molecule-Text Alignment

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This paper studies the problem of molecule-text alignment, which aims to project molecules and their textual descriptions into a joint latent space for downstream tasks including molecule search and molecular property prediction. Previous approaches typically combine graph structure mining with contrastive learning to enhance joint representation learning. However, they typically neglect fine-grained semantic relationships between substructures and texts, leading to suboptimal performance on downstream tasks. Towards this end, we propose a novel approach named Retrieval-guided Twin Fusion with Similarity-aware Contrast (RISEN) for molecule-text alignment. The core idea of RISEN is to construct a latent twin molecule for each substructure with cross-modal retrieval for semantic enhancement. In particular, for each substructure query, we retrieve relevant textual descriptions and sample several molecules that share similar descriptions of substructures. Then, we aggregate their representations via attention pooling for a twin latent representation, which would be further fused with the original substructure for representation enrichment. In addition, we measure the similarity across substructures and texts, which would further guide cross-modal contrastive learning with soft thresholding. Extensive experiments on benchmark datasets validate the superiority of the proposed RISEN in comparison with existing baselines.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Information systems** → **Information retrieval**; • **Applied computing** → **Bioinformatics**.

Additional Key Words and Phrases: molecule-text alignment, contrastive learning, retrieval-augmented learning, molecular representation learning

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1 Introduction

Molecules encode rich chemical information whose systematic understanding is central to drug discovery, materials science, and molecular property prediction [14, 23, 27]. Machine learning has emerged as a powerful lens for this task, with language models offering a particularly promising route, as natural-language descriptions of molecules provide functional and contextual cues that complement structural representations [9]. Molecule-text alignment has thus become a central paradigm [4], enabling models to jointly embed molecular structures and textual descriptions into a shared semantic space for cross-modal reasoning.

Most prior work on molecule-text alignment follows the contrastive representation learning paradigm. Early approaches such as Text2Mol [4] and MoMu [19] learned aligned embeddings from paired molecules and textual descriptions to support text-to-molecule retrieval. Building on this, KV-PLM [25] bridged molecules and biomedical text through unified language modeling, while MoleculeSTM [9] performed large-scale structure-text contrastive

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pre-training to obtain more general cross-modal representations that transfer to multiple downstream tasks. More recent methods pursue finer-grained correspondences. MolCA [10] introduces a cross-modal projector to bridge graph and language spaces, while MolBridge [14] aligns molecular substructures with chemical phrases, further improving cross-modal learning by injecting local structural supervision.

Despite this progress, two limitations remain. **Firstly**, finer-grained matching can amplify noise. Real-world descriptions are often incomplete, ambiguous, and context-dependent. Expanding supervision from instance-level pairs to many substructure–phrase pairs can introduce weak or incorrect alignments, diluting useful signals and degrading generalization [5, 8]. **Secondly**, most methods rely on hard contrastive objectives with one-hot targets, treating all other in-batch samples as negatives [3]. However, molecule–text relevance is often inherently soft, given that many molecules share scaffolds, functional groups, or pharmacophores, and treating them as negatives introduces false negatives that distort semantic neighborhoods. Soft contrastive targets derived from embedding similarities have shown promise in mitigating this issue in other alignment settings [15].

We propose RISEN, a retrieval-guided molecule–text alignment framework designed to address both limitations. To enrich substructure representations with broader chemical context that isolated fragment learning alone cannot provide, Retrieval-guided Twin Fusion retrieves semantically related mirror molecules for each substructure and aggregates them into an attention-pooled twin prototype, which is then fused with the original embedding to enable local motifs to inherit structural context from complete molecules. To mitigate false negatives arising from chemically similar in-batch molecules, Similarity-aware Contrast replaces one-hot targets with similarity-based soft targets derived from MoLFormer [18] embeddings, better preserving semantic neighborhoods in the embedding space. Across molecule–text retrieval and molecular property prediction benchmarks, RISEN achieves consistent overall gains over strong baselines.

Our contributions are summarized as follows: ❶ We present a retrieval-augmented perspective for substructure-level molecule–text alignment, where semantically related complete molecules are leveraged to enrich local fragment representations with broader chemical context. ❷ We propose RISEN, a retrieval-guided framework that enriches substructure representations via twin prototype fusion and alleviates false negatives through similarity-aware soft targets. ❸ Extensive experiments on molecule–text retrieval and molecular property prediction benchmarks demonstrate consistent gains over strong baselines across both tasks.

2 Related Work

Molecule–Text Multimodal Modeling. Molecule–text alignment methods can be categorized by alignment granularity. Global-level methods align complete molecular structures with full descriptions via contrastive learning. MoMu [19] and MoleculeSTM [9] apply large-scale contrastive pre-training, MolCA [10] introduces a cross-modal projector to bridge graph and language encoders, and instruction-tuned models such as InstructMol [2] and BioT5+ [16] leverage multi-task learning for broader generalization. These methods operate on global molecule–text pairs and overlook fine-grained structural correspondences. Fine-grained methods instead capture local correspondences between molecular fragments and chemical phrases. Yu et al. [24] explored modality blending to encourage multi-scale representations, Min et al. [13] applied optimal transport to align atom, motif, and global levels, and Atomas [27] designed a hierarchical adaptive alignment model. MolBridge [14] constructs explicit substructure–phrase pairs via chemical fragmentation tools and applies a self-refinement mechanism to filter noisy alignments. Unlike these approaches, RISEN enriches substructure representations with semantically related complete molecules, providing broader chemical context without requiring explicit fragment-level annotations.

Contrastive Learning with Soft Targets. Standard contrastive objectives treat all non-matching in-batch samples as negatives, creating false negatives when semantically similar samples co-occur [3]. Li et al. [8] proposed momentum-distilled soft labels to handle noisy correspondences in vision-language pre-training, and Park et al. [15] derived soft targets from embedding similarities to preserve semantic neighborhoods in multilingual alignment. In the molecular domain, Wang et al. [22] demonstrated that structurally similar molecules cause harmful false negatives and proposed fragment contrast as mitigation. Our Similarity-aware Contrast extends this line by constructing soft contrastive targets from pairwise MoLFormer [18] embedding similarities, directly reducing the penalty on chemically similar in-batch molecules.

Retrieval-Augmented Learning. Retrieval augmentation has been applied across NLP and vision-language modeling to enrich representations with relevant external knowledge [6]. In the molecular domain, retrieval has primarily been used at the input level for generation tasks such as molecule captioning [7]. RISEN differs by applying retrieval at the representation level during training, aggregating retrieved mirror molecules into an attention-pooled prototype fused with the original substructure embedding, without modifying the inference pipeline.

3 Methodology

Framework Overview. In this paper, we introduce a retrieval-augmented alignment framework that extends substructure-aware contrastive learning with structured retrieval and soft supervision. The core idea is to enrich isolated substructure representations with chemical context from semantically related complete molecules, while replacing hard contrastive targets with similarity-aware soft labels to mitigate false negatives.

3.1 Retrieval-guided Twin Fusion

Substructure representations learned in isolation often lack the broader molecular context needed to support accurate cross-modal alignment [26, 28]. To address this, we retrieve semantically related complete molecules for each substructure query, aggregate their embeddings into an attention-pooled twin prototype via an attention-based mechanism, and fuse it with the original substructure representation to provide richer chemical context.

For each substructure query, we collect the textual descriptions aligned with it in the training set, and take all complete molecules aligned with any of these descriptions as its mirror set. The mirror map $\mathcal{M}$ is precomputed before training, and at each step we sample $k \leq 5$ mirror molecules uniformly at random.

Specifically, we aggregate the embeddings of k retrieved mirror molecules $\{\mathbf{m}_1, \dots, \mathbf{m}_k\}$ into a prototype $\mathbf{p}$ as follows:

$$w_i = \frac{\exp(\mathbf{h}^\top \mathbf{m}_i / \sqrt{d})}{\sum_{j=1}^k \exp(\mathbf{h}^\top \mathbf{m}_j / \sqrt{d})}, \quad \mathbf{p} = \frac{\sum_{i=1}^k w_i \mathbf{m}_i}{\left\| \sum_{i=1}^k w_i \mathbf{m}_i \right\|_2}. \quad (1)$$

The original representation $\mathbf{h}$ is then enriched with this prototype information:

$$\mathbf{h}' = \text{Norm}((1 - \alpha)\mathbf{h} + \alpha\mathbf{p}), \quad (2)$$

where d is the embedding dimension and α is a fusion coefficient. This design enriches substructure representations with broader chemical context.

3.2 Similarity-aware Contrastive Learning

Hard contrastive objectives treat all non-matching in-batch samples as negatives [3, 21], yet many molecules share scaffolds, functional groups, or pharmacophores, creating harmful false negatives. To address this, we leverage pairwise

MolFormer embedding similarities to construct soft contrastive targets that reduce the penalty on chemically similar in-batch molecules, better preserving semantic neighborhoods during training.

Specifically, let S_{ij} denote the cosine similarity between the normalized MolFormer embeddings of molecules i and j in a mini-batch. We apply a threshold τ to filter weak similarities and derive a soft target distribution $\mathbf{q}^{\text{soft}}$ via a temperature-scaled softmax:

$$\tilde{S}_{ij} = S_{ij} \mathbb{I}(S_{ij} \geq \tau), \quad q_{ij}^{\text{soft}} = \frac{\exp(\tilde{S}_{ij}/T)}{\sum_{l=1}^B \exp(\tilde{S}_{il}/T)}. \quad (3)$$

For each retrieval direction, the final target distribution $\mathbf{q}$ is a mixture of hard labels $\mathbf{q}_{\text{hard}}$ and soft labels $\mathbf{q}_{\text{soft}}$ controlled by a mixing weight β :

$$\mathbf{q} = (1 - \beta) \mathbf{q}_{\text{hard}} + \beta \mathbf{q}_{\text{soft}}. \quad (4)$$

This design yields a smoother supervision signal that better preserves semantic neighborhoods.

3.3 Optimization Objectives

Effective cross-modal alignment benefits from supervision that is both directionally balanced and robust to noise [8]. To this end, we optimize RISEN with two complementary objectives: a bidirectional contrastive loss on soft-hard mixed targets to align molecule and text representations, and a prototype contrastive loss to enforce alignment consistency between the fused representation and the twin prototype.

Bidirectional Contrastive Loss with Soft Labels. Given a mini-batch of molecule-side inputs x_m^i and text-side inputs x_t^j , we first encode them into feature representations $u_i = f_m(x_m^i)$ and $v_j = f_t(x_t^j)$. We define the pairwise similarity $\sigma_{i,j}(u, v)$ as:

$$\sigma_{i,j}(u, v) = \exp\left(\gamma \cdot \frac{u_i^T v_j}{\|u_i\|_2 \|v_j\|_2}\right), \quad (5)$$

where γ is a learnable logit scale. After masking invalid pair types, and using direction-specific mixed target distributions $q_{ij}^{s \rightarrow t}$ and $q_{ij}^{t \rightarrow s}$, the bidirectional losses are defined as:

$$\mathcal{L}_{s \rightarrow t} = -\frac{1}{B} \sum_{i=1}^B \sum_{j=1}^B q_{ij}^{s \rightarrow t} \log \frac{\sigma_{i,j}}{\sum_{l=1}^B \sigma_{i,l}}, \quad (6)$$

$$\mathcal{L}_{t \rightarrow s} = -\frac{1}{B} \sum_{i=1}^B \sum_{j=1}^B q_{ij}^{t \rightarrow s} \log \frac{\sigma_{i,j}}{\sum_{l=1}^B \sigma_{l,j}}. \quad (7)$$

The contrastive loss in this branch is the average of both directions: $\mathcal{L}_{\text{cont}} = \frac{1}{2} (\mathcal{L}_{s \rightarrow t} + \mathcal{L}_{t \rightarrow s})$.

Total Loss. We optimize the contrastive loss on both the fused substructure representation $\mathbf{h}'$ and the mirror molecule prototype $\mathbf{p}$, with an additional classification loss $\mathcal{L}_{\text{class}}$ following the self-refinement protocol of MolBridge [14] to iteratively filter unreliable alignment pairs:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{cont}}(\mathbf{h}', \mathbf{t}) + \lambda \cdot \mathcal{L}_{\text{cont}}(\mathbf{p}, \mathbf{t})}_{\mathcal{L}_{\text{combined}}} + \mathcal{L}_{\text{class}}, \quad (8)$$

where λ is the prototype loss weight and $\mathbf{t}$ is the text representation. Together, these objectives encourage RISEN to learn representations that are semantically aligned, structurally enriched, and robust to noisy correspondences.

Table 1. Performance comparison for zero-shot molecule-text retrieval on PCDes test set (scaffold split). Baseline results are quoted from Park et al. [14]. We **bold** the best results and underline the second-best.

Methods	# Params	Text to Molecule				Molecule to Text			
		R@1	R@5	R@10	MRR	R@1	R@5	R@10	MRR
1D SMILES + 2D Graph									
MoMu [19]	111M	4.90	14.48	20.69	10.33	5.08	12.82	18.93	9.89
MolFM [11]	138M	16.14	30.67	39.54	23.63	13.90	28.69	36.21	21.42
MolFM (fine-tuned) [11]	138M	29.39	50.26	58.49	39.34	29.76	50.53	58.63	39.56
MolCA [10]	111M	35.09	62.14	69.77	47.33	37.95	66.81	74.48	50.80
1D SMILES									
MoleculeSTM [9]	120M	35.80	–	–	–	39.50	–	–	–
Atomas-base [27]	271M	39.08	59.72	66.56	47.33	37.88	59.22	65.56	47.81
Atomas-large [27]	825M	49.08	68.32	73.16	57.79	46.22	66.02	72.32	55.52
MolBridge [14]	155M	50.45	70.83	76.11	59.63	52.76	73.54	78.55	62.25
RISEN	155M	52.05	73.47	78.72	61.56	55.40	75.04	80.09	64.28

Table 2. Performance comparison for zero-shot molecule-text retrieval on PubChem324kV2 test set. Baseline results are quoted from Park et al. [14]. We **bold** the best results and underline the second-best.

Methods	Text to Molecule		Molecule to Text	
	R@1	R@20	R@1	R@20
1D SMILES + 2D Graph				
MoMu-S [19]	40.8	86.1	40.9	86.2
MoMu-K [19]	41.6	87.8	41.8	87.5
MoleculeSTM [9]	44.3	90.3	45.8	88.4
MolCA [10]	66.0	93.5	66.6	94.6
1D SMILES				
SciBERT [1]	37.5	85.2	39.7	85.8
KV-PLM [25]	37.7	85.5	38.8	86.0
MolBridge [14]	<u>70.9</u>	<u>95.6</u>	<u>75.0</u>	<u>97.4</u>
RISEN	73.7	96.2	77.2	97.1

4 Experiments

4.1 Experimental Settings

Datasets. Following Park et al. [14], we train on 431,877 molecule–description pairs augmented to 2M via substructure-level alignment, and evaluate on PCDes [25] (scaffold split) and PubChem324kV2 [9] for retrieval, and eight MoleculeNet [23] datasets for property prediction.

Metrics. We report Recall@1/5/10/20 [12] and MRR [20] for zero-shot molecule-text retrieval. For molecular property prediction, we report ROC-AUC (%).

Baselines. We compare against representative multimodal methods, including MoMu [19], MolFM [11], MolCA [10], KV-PLM [25], MoleculeSTM [9], SciBERT [1], Atomas [27], and MolBridge [14]. The reported baseline results are taken from Park et al. [14].

Table 3. Results for molecular property prediction tasks (ROC-AUC %) on the MoleculeNet benchmark. We **bold** the best results and underline the second-best.

Methods	BBBP	Tox21	ToxCast	ClinTox	MUV	HIV	BACE	SIDER	Avg.
MoleculeSTM [9]	70.6	75.7	65.2	86.6	65.7	77.0	82.0	63.7	73.3
MolFM [11]	72.9	77.2	64.4	79.7	76.0	78.8	83.9	64.2	74.6
MoMu [19]	70.5	75.6	63.4	79.9	70.6	75.9	76.7	60.5	71.6
MolCA-SMILES [10]	70.8	76.0	56.2	89.0	—	—	79.3	61.1	—
Atomax [27]	73.7	77.9	66.9	93.2	76.3	80.6	83.1	64.4	77.0
MolBridge [14]	<u>77.6</u>	84.7	70.3	94.8	76.8	77.8	<u>84.5</u>	<u>66.9</u>	<u>79.2</u>
RISEN	78.5	<u>78.4</u>	74.5	95.3	80.0	<u>80.0</u>	87.8	67.7	80.3

Table 4. Ablation study on PubChem324kV2. SL: Soft Labels, AP: Attention Pooling, ML: Mirror Loss, PF: Prototype Fusion. **Bold** = best; underline = second-best.

Methods	Text to Molecule				Molecule to Text			
	R@1	R@5	R@10	MRR	R@1	R@5	R@10	MRR
RISEN	73.65	91.50	94.45	<u>81.72</u>	77.15	93.15	95.75	84.45
RISEN w/o SL	72.85	91.50	94.15	80.99	76.65	<u>92.90</u>	<u>95.60</u>	<u>84.06</u>
RISEN w/o AP	73.25	90.65	93.60	81.06	<u>76.90</u>	92.35	94.85	83.78
RISEN w/o ML	<u>73.70</u>	90.80	93.80	81.33	76.00	93.15	95.35	83.60
RISEN w/o PF	74.30	<u>91.15</u>	<u>94.35</u>	81.98	76.45	92.80	95.20	83.84
MolBridge	70.90	—	—	—	75.00	—	—	—

4.2 Empirical Results

Zero-shot Molecule-Text Retrieval. Table 1 and Table 2 summarize results on PCDes and PubChem324kV2, respectively. RISEN achieves the best performance on most metrics across both benchmarks and retrieval directions, with particularly strong gains in M2T. These results demonstrate the benefit of retrieval-enriched substructure representations over isolated fragment learning.

Molecular Property Prediction. Following Park et al. [14], we evaluate on eight MoleculeNet [23] classification datasets using the scaffold split from DeepChem [17]. Table 3 shows that RISEN achieves the best average ROC-AUC with gains on most tasks, confirming effective transfer to downstream property-relevant chemical semantics.

Ablation Study. To assess the contribution of each component, we conduct an ablation study on PubChem324kV2 (Table 4). Removing soft labels clearly hurts T2M R@1 and MRR, supporting their role in mitigating false negatives. Attention pooling improves both retrieval directions. Removing mirror loss or prototype fusion yields small T2M gains but degrades M2T, suggesting that these components mainly stabilize M2T alignment.

Sensitivity Analysis. We vary the prototype fusion weight α and the soft-label mixing weight β on PCDes. As shown in Figure 1, $\alpha=0.1$ provides the best balance across directions, indicating that a small amount of prototype fusion suffices; $\beta=0.3$ yields the strongest results, while the sharp drop at $\beta=0.8$ confirms that soft labels should supplement rather than replace the original objective.

Case Study. Figure 2 compares functional-group phrases identified by RISEN and MolBridge [14] against PubChem ground truth. RISEN recovers a more complete set, suggesting that retrieval-enriched substructure representations yield more faithful motif-level alignment.

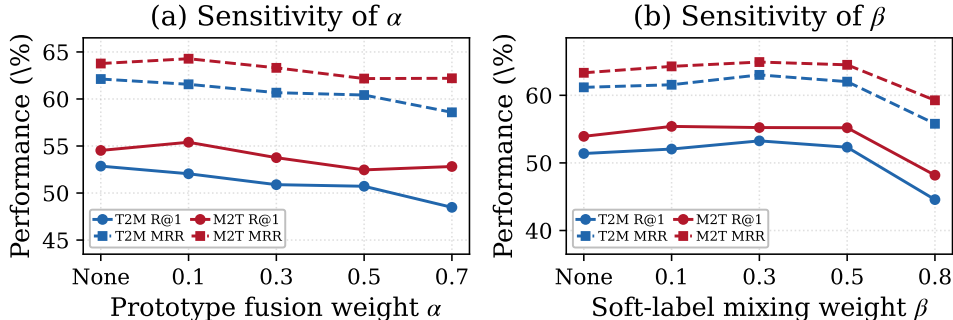


Fig. 1. Sensitivity analysis of the prototype fusion weight α (left) and the soft-label mixing weight β (right) on PCDes.

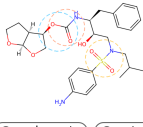
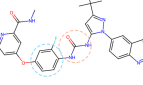
Molecule	Phrases found from PubChem	Phrases found by RISEN	Phrases found by molbridge
 carbamate ester sulfonamide	Amprenavir is an antiviral drug. It is a furofuran, a carbamate ester and a sulfonamide . It is a human immunodeficiency virus type-1 protease inhibitor used in HIV treatment.	Amprenavir is an antiviral drug. It is a furofuran, a carbamate ester and a sulfonamide . It is a human immunodeficiency virus type-1 protease inhibitor used in HIV treatment.	Amprenavir is an antiviral drug. It is a furofuran, a carbamate ester and a sulfonamide . It is a human immunodeficiency virus type-1 protease inhibitor used in HIV treatment.
 urea phenyl	A compound of the class of ureas that is urea in which one nitrogen bears a 3-tert-butyl-1-(quinolin-6-yl)-1H-pyrazol-5-yl substituent, while the other bears a 2-fluoro-4-([2-(methylcarbamoyl)pyridin-4-yl]oxy) phenyl substituent. It has a role as an antineoplastic agent and receptor tyrosine kinase inhibitor.	A compound of the class of ureas that is urea in which one nitrogen bears a 3-tert-butyl-1-(quinolin-6-yl)-1H-pyrazol-5-yl substituent, while the other bears a 2-fluoro-4-([2-(methylcarbamoyl)pyridin-4-yl]oxy) phenyl substituent. It has a role as an antineoplastic agent and receptor tyrosine kinase inhibitor.	A compound of the class of ureas that is urea in which one nitrogen bears a 3-tert-butyl-1-(quinolin-6-yl)-1H-pyrazol-5-yl substituent, while the other bears a 2-fluoro-4-([2-(methylcarbamoyl)pyridin-4-yl]oxy) phenyl substituent. It has a role as an antineoplastic agent and receptor tyrosine kinase inhibitor.

Fig. 2. Qualitative comparison of functional-group phrases identified by RISEN and MolBridge [14] against ground-truth phrases from PubChem descriptions.

5 Conclusion

We present RISEN, a retrieval-guided framework for molecule-text alignment that effectively enriches substructure representations via twin prototype fusion and alleviates false negatives through similarity-aware soft targets. Experiments on zero-shot retrieval and molecular property prediction benchmarks show consistent overall gains over strong baselines, highlighting the effectiveness of semantic retrieval and soft contrastive supervision in cross-modal molecular alignment.

6 Limitations

As with most prototype-based contrastive learning approaches, our method encodes additional mirror molecules during training, which moderately increases memory usage. This cost is bounded (at most $k=5$ per substructure sample) and enables structurally enriched representations that we find empirically beneficial, though scaling to substantially larger batch sizes or longer retrieval neighborhoods may pose challenges. Additionally, our soft-label mechanism relies on similarities computed in the frozen MoLFormer embedding space, which may not fully capture all task-relevant chemical semantics, potentially limiting soft-label quality for underrepresented molecular scaffolds or domain-shifted distributions. Finally, our evaluation focuses on molecule-text retrieval and molecular property prediction, and extending

RISEN to generative tasks such as molecule captioning or text-guided molecule generation would provide a more complete picture of its general utility.

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